Daily Literature Review — 2026-10-31

Daily Literature Review: AI and machine learning in protein function prediction

Date: 2026-10-31

Research Question

How are machine-learning methods being used to predict protein function, what datasets and evaluation metrics are used, and what limitations remain?

Summary

This document compiles 40 recent papers related to the topic. AI drafting was unavailable, so the content below is built directly from bibliographic metadata. Each entry includes title, authors, year, source, DOI/URL, and a reminder to extract findings from the original paper. This is an evidence-collection scaffold, not a finished scientific review.

Evidence Table (Bibliographic)

2026

Rapid directed evolution guided by protein language models and epistatic interactions.

- Authors: Tran VQ, Nemeth M, Bartie LJ, Chandrasekaran SS, Fanton A, Moon HC, Hie BL, Konermann S

- Source: PubMed

- PMID: 41712694

- Link: https://pubmed.ncbi.nlm.nih.gov/41712694/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Decoding and deciphering a subcellular ZIP code system.

- Authors: Ille AM, Markosian C, Pasqualini R, Arap W

- Source: PubMed

- PMID: 41820158

- Link: https://pubmed.ncbi.nlm.nih.gov/41820158/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Discovery of natural RORγt inhibitor using machine learning, virtual screening, and in vivo validation.

- Authors: Yoo H, Han SJ, Lee JE, Cho C, Hong D, Jeong B, Kim S, Jung GY

- Source: PubMed

- PMID: 40915561

- Link: https://pubmed.ncbi.nlm.nih.gov/40915561/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Systematic Multi-Target QSAR Benchmarking: Machine Learning Algorithms, Molecular Descriptors, and Validation.

- Authors: Alshehade SA, Al Jabal G, Jebril IH

- Source: PubMed

- PMID: 42300227

- Link: https://pubmed.ncbi.nlm.nih.gov/42300227/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine Learning for Studying Plant Protein Function and Evolution

- Authors: Ankan Das, Dibyendu Seth, Bodhisattya Pal, Nidhi, Sandip Debnath

- Source: OpenAlex

- DOI: 10.1201/9781003545781-20

- Link: https://doi.org/10.1201/9781003545781-20

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Leveraging Gemma 4 Large Language Model for Protein Function Prediction and Interpretability Application of AI Models for Protein Function Prediction from Amino Acid Sequences

- Authors: Trinh Quang Minh, Ngo Thi Lan

- Source: OpenAlex

- DOI: 10.38124/ijisrt/26apr247

- Link: https://doi.org/10.38124/ijisrt/26apr247

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2025

Can AI-Predicted Complexes Teach Machine Learning to Compute Drug Binding Affinity?

- Authors: Hsu WT, Grevtsev S, Herz AM, Douglas T, Magarkar A, Biggin PC

- Source: PubMed

- PMID: 41369266

- Link: https://pubmed.ncbi.nlm.nih.gov/41369266/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Deciphering the mechanism of baicalein in cervical cancer via bioinformatics, machine learning and computational simulations: PIM1 and CDK2 are key target proteins.

- Authors: Wang S, Liu C, Ye D, Qi J, Xing Y, Wang J, Fan X, Li X

- Source: PubMed

- PMID: 40339869

- Link: https://pubmed.ncbi.nlm.nih.gov/40339869/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

LCN2 drives ferroptosis-associated ischemia-reperfusion injury after renal transplantation: integrated machine learning and in vivo validation.

- Authors: Wu Z, Yu B, He Q, Huang C

- Source: PubMed

- PMID: 41205031

- Link: https://pubmed.ncbi.nlm.nih.gov/41205031/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine Learning Prediction of 90-Day Mortality in HBV-Related ACLF Using Olink-Derived Inflammatory Protein Signatures.

- Authors: Zhang Y, Sun L, Liu H, Shi K, Lu Y, Feng Y, Wang X

- Source: PubMed

- PMID: 41255250

- Link: https://pubmed.ncbi.nlm.nih.gov/41255250/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine learning driven prediction of drug efficacy in lung cancer: based on protein biomarkers and clinical features.

- Authors: Li J, Chen A, Liu Z, Wei S, Zhang J, Chen J, Shi C

- Source: PubMed

- PMID: 40355026

- Link: https://pubmed.ncbi.nlm.nih.gov/40355026/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Applications of Machine Learning Approaches for the Discovery of SARS-CoV-2 PLpro Inhibitors.

- Authors: Pal S, Nance KD, Joshi DR, Kales SC, Ye L, Hu X, Shamim K, Zakharov AV

- Source: PubMed

- PMID: 39818814

- Link: https://pubmed.ncbi.nlm.nih.gov/39818814/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

AI and Machine Learning in Biology: From Genes to Proteins

- Authors: Zaw Myo Hein, Dhanyashri Guruparan, Blaire Okunsai, Che Mohd Nasril Che Mohd Nassir, Muhammad Danial Che Ramli, Suresh Kumar

- Source: OpenAlex

- DOI: 10.3390/biology14101453

- Link: https://doi.org/10.3390/biology14101453

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

A REVIEW ON EXPLAINABLE AI IN ProTEIN FUNCTION

- Authors: Aastha Katiyar

- Source: OpenAlex

- DOI: 10.22541/au.174912536.68729526/v1

- Link: https://doi.org/10.22541/au.174912536.68729526/v1

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

AI-Driven Functional Protein Design: Machine Learning Approaches for Screening and Optimization

- Authors: Yunze Jiang

- Source: OpenAlex

- DOI: 10.1145/3778450.3778490

- Link: https://doi.org/10.1145/3778450.3778490

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2024

AI-Driven Deep Learning Techniques in Protein Structure Prediction.

- Authors: Chen L, Li Q, Nasif KFA, Xie Y, Deng B, Niu S, Pouriyeh S, Dai Z

- Source: PubMed

- PMID: 39125995

- Link: https://pubmed.ncbi.nlm.nih.gov/39125995/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine learning-based reproducible prediction of type 2 diabetes subtypes.

- Authors: Tanabe H, Sato M, Miyake A, Shimajiri Y, Ojima T, Narita A, Saito H, Tanaka K

- Source: PubMed

- PMID: 39168869

- Link: https://pubmed.ncbi.nlm.nih.gov/39168869/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine learning interpretable models of cell mechanics from protein images.

- Authors: Schmitt MS, Colen J, Sala S, Devany J, Seetharaman S, Caillier A, Gardel ML, Oakes PW

- Source: PubMed

- PMID: 38194965

- Link: https://pubmed.ncbi.nlm.nih.gov/38194965/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

AKT and EZH2 inhibitors kill TNBCs by hijacking mechanisms of involution.

- Authors: Schade AE, Perurena N, Yang Y, Rodriguez CL, Krishnan A, Gardner A, Loi P, Xu Y

- Source: PubMed

- PMID: 39385030

- Link: https://pubmed.ncbi.nlm.nih.gov/39385030/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine Learning Methods in Protein-Protein Docking.

- Authors: Michalik I, Kuder KJ

- Source: PubMed

- PMID: 38987466

- Link: https://pubmed.ncbi.nlm.nih.gov/38987466/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Enhancing the Predictive Power of Machine Learning Models through a Chemical Space Complementary DEL Screening Strategy.

- Authors: Suo Y, Qian X, Xiong Z, Liu X, Wang C, Mu B, Wu X, Lu W

- Source: PubMed

- PMID: 39441849

- Link: https://pubmed.ncbi.nlm.nih.gov/39441849/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Deep learning methods for protein function prediction

- Authors: Frimpong Boadu, Ahhyun Lee, Jianlin Jack Cheng

- Source: OpenAlex

- DOI: 10.1002/pmic.202300471

- Link: https://doi.org/10.1002/pmic.202300471

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

AI-Driven Deep Learning Techniques in Protein Structure Prediction

- Authors: Lingtao Chen, Qiaomu Li, Kazi Fahim Ahmad Nasif, Ying Xie, Bobin Deng, Shuteng Niu, Seyedamin Pouriyeh, Zhiyu Dai

- Source: OpenAlex

- DOI: 10.3390/ijms25158426

- Link: https://doi.org/10.3390/ijms25158426

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine Learning-Assisted Protein Structure Prediction: An AI Approach for Biochemical Insights

- Authors: Radha Mahendran, Akriti Dogra, Dinesh Mendhe, S. B G Tilak Babu, Shriniket Dixit, Surya Pratap Singh

- Source: OpenAlex

- DOI: 10.1109/iconstem60960.2024.10568895

- Link: https://doi.org/10.1109/iconstem60960.2024.10568895

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Protein function prediction as approximate semantic entailment

- Authors: Maxat Kulmanov, Francisco J. Guzmán■Vega, Paula D Duek, Lydie Lane, Stefan T. Arold, Robert Hoehndorf

- Source: OpenAlex

- DOI: 10.1038/s42256-024-00795-w

- Link: https://doi.org/10.1038/s42256-024-00795-w

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Advances in AI and machine learning for predictive medicine

- Authors: Alok K. Sharma, Artem Lysenko, Shangru Jia, Keith A. Boroevich, Tatsuhiko Tsunoda

- Source: OpenAlex

- DOI: 10.1038/s10038-024-01231-y

- Link: https://doi.org/10.1038/s10038-024-01231-y

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Improving protein function prediction by learning and integrating representations of protein sequences and function labels

- Authors: Frimpong Boadu, Jianlin Jack Cheng

- Source: OpenAlex

- DOI: 10.1093/bioadv/vbae120

- Link: https://doi.org/10.1093/bioadv/vbae120

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Leveraging machine learning models for peptide–protein interaction prediction

- Authors: Yin Song, Xuenan Mi, Diwakar Shukla

- Source: OpenAlex

- DOI: 10.1039/d3cb00208j

- Link: https://doi.org/10.1039/d3cb00208j

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2023

Novel machine learning approaches revolutionize protein knowledge.

- Authors: Bordin N, Dallago C, Heinzinger M, Kim S, Littmann M, Rauer C, Steinegger M, Rost B

- Source: PubMed

- PMID: 36504138

- Link: https://pubmed.ncbi.nlm.nih.gov/36504138/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Hierarchical graph learning for protein-protein interaction.

- Authors: Gao Z, Jiang C, Zhang J, Jiang X, Li L, Zhao P, Yang H, Huang Y

- Source: PubMed

- PMID: 36841846

- Link: https://pubmed.ncbi.nlm.nih.gov/36841846/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine Learning-Guided Protein Engineering

- Authors: Petr Kouba, Pavel Kohout, Faraneh Haddadi, Anton Bushuiev, Raman Samusevich, Ji■í Sedlá■, Ji■■■ Damborský, Tomáš Pluskal

- Source: OpenAlex

- DOI: 10.1021/acscatal.3c02743

- Link: https://doi.org/10.1021/acscatal.3c02743

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2022

Machine Learning Approaches for Metalloproteins.

- Authors: Yu Y, Wang R, Teo RD

- Source: PubMed

- PMID: 35209064

- Link: https://pubmed.ncbi.nlm.nih.gov/35209064/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

AI-Based Protein Structure Prediction in Drug Discovery: Impacts and Challenges.

- Authors: Schauperl M, Denny RA

- Source: PubMed

- PMID: 35727311

- Link: https://pubmed.ncbi.nlm.nih.gov/35727311/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Protein–protein interaction prediction with deep learning: A comprehensive review

- Authors: Farzan Soleymani, Eric Paquet, Herna Lydia Viktor, Wojtek J. Michalowski, Davide Spinello

- Source: OpenAlex

- DOI: 10.1016/j.csbj.2022.08.070

- Link: https://doi.org/10.1016/j.csbj.2022.08.070

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Protein Function Analysis through Machine Learning

- Authors: Chris Avery, John Patterson, Tyler J. Grear, Theodore D. Frater, Donald J. Jacobs

- Source: OpenAlex

- DOI: 10.3390/biom12091246

- Link: https://doi.org/10.3390/biom12091246

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Dowker complex based machine learning (DCML) models for protein-ligand binding affinity prediction

- Authors: Xiang Liu, Huitao Feng, Jie Wu, Kelin Xia

- Source: OpenAlex

- DOI: 10.1371/journal.pcbi.1009943

- Link: https://doi.org/10.1371/journal.pcbi.1009943

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

XGboost-Ampy: Identification of AMPylation Protein Function Prediction Using Machine Learning

- Authors: Zar Nawab Khan Swati, Ali Ghulam, Muhammad Sohail, Jawad Usman Arshed, Rahu Sikander, Muhammad Shahid Malik, Nauman Khan

- Source: OpenAlex

- DOI: 10.21015/vtcs.v10i2.1347

- Link: https://doi.org/10.21015/vtcs.v10i2.1347

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Prediction of Diabetes Empowered With Fused Machine Learning

- Authors: Usama Ahmed, Ghassan F. Issa, Muhammad Adnan Khan, Shabib Aftab, Muhammad Farhan Khan, Raed A. Said, Taher M. Ghazal, Munir Ahmad

- Source: OpenAlex

- DOI: 10.1109/access.2022.3142097

- Link: https://doi.org/10.1109/access.2022.3142097

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Hom-Complex-Based Machine Learning (HCML) for the Prediction of Protein–Protein Binding Affinity Changes upon Mutation

- Authors: Xiang Liu, Huitao Feng, Jie Wu, Kelin Xia

- Source: OpenAlex

- DOI: 10.1021/acs.jcim.2c00580

- Link: https://doi.org/10.1021/acs.jcim.2c00580

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2021

Structure-based protein function prediction using graph convolutional networks

- Authors: Vladimir Gligorijevi■, P. Douglas Renfrew, Tomasz Ko■ció■ek, Julia Koehler Leman, Daniel Berenberg, Tommi Vatanen, Chris Chandler, Bryn C. Taylor

- Source: OpenAlex

- DOI: 10.1038/s41467-021-23303-9

- Link: https://doi.org/10.1038/s41467-021-23303-9

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Recommended Next Steps

1. Open the attached individual PDFs (or follow the DOI/URL links).

2. For each relevant paper, extract: main method, datasets used, evaluation metrics, key quantitative results, and stated limitations.

3. Record contradictions or gaps across papers.

4. Only after verification, move evidence into a formal manuscript.

Scientific Rule

AI is an assistant, not a source. Never submit an AI-generated or placeholder claim without checking the original paper.

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